LPA-MNI+ML

A multi-level variant of LPA-MNI for community detection: what it is, how it behaves, and what the current results do and do not show

1. Summary

LPA-MNI+ML is a community detection algorithm built in this project. It keeps the second phase of the LPA-MNI algorithm (Li et al., Entropy 2021) - label propagation with degree-centrality tie-breaking - but replaces the first phase, a single flat greedy modularity pass, with Louvain-style multi-level coarsening. The purpose is to let the algorithm merge whole communities, which the flat pass cannot do.

- **Modularity.** Multi-level seeding raises modularity on every dataset compared with LPA-MNI, by a large margin on Cora (0.726 to 0.816), Florentine Families (0.289 to 0.391) and American College Football (0.557 to 0.605) and Yeast (0.727 to 0.818). Among the five algorithms compared here it has the highest modularity on 6 of 6 datasets.
- **NMI.** The effect is mixed. NMI rises on Florentine, Dolphins and Cora, rises slightly on Yeast, is about flat on Football, and falls on Karate Club (0.837 to 0.691).
- **Scope.** This report compares five label-propagation-family algorithms only (LPA, LPA-MNI, FLPA, LPA-S, LPA-MNI+ML). Other families, such as modularity optimizers, are not part of this comparison, so 'best' in this report means best of these five, not best overall.
- **Caveat.** All numbers come from one run per algorithm and dataset, on six datasets, three of which (Florentine Families, Dolphins and Yeast) use proxy ground truth. Differences of about 0.01 to 0.02 are within run-to-run noise and should not be read as rankings.

2. Background

2.1 Community detection and the algorithms compared

Community detection partitions a graph's nodes into groups that are densely connected internally and sparsely connected to the rest. Every algorithm compared in this report belongs to the label-propagation family: there is no global objective, and each node repeatedly adopts a label from its neighbors until labels stop changing. They differ in how a node picks its label and in the order nodes are visited.

- **LPA.** Plain label propagation. Fast, but random, and it can collapse or fragment.
- **FLPA.** A queue-based variant that only revisits nodes whose neighborhood changed.
- **LPA-S.** A deterministic variant that weights each neighbor's vote by a similarity score.
- **LPA-MNI.** Seeds the labels with a greedy modularity pass, then propagates (described below).
- **LPA-MNI+ML.** The algorithm this report is about: LPA-MNI with a multi-level seeding step.

2.2 LPA-MNI (the original)

LPA-MNI has two phases. **Phase 1** visits nodes in descending degree order and moves each node into the neighboring community that gives the largest modularity gain, repeating until a full pass moves nobody. **Phase 2** uses the resulting communities as initial labels and runs asynchronous label propagation in the same fixed order. Each node takes the most common label among its neighbors. A tie is broken by choosing the label of the tied neighbor with the highest degree centrality, DC = k / (n - 1), so no randomness is involved.

Phase 1 has a structural limit. A node can only join the community of one of its neighbors, so two existing communities can never merge. That puts a ceiling on the modularity the algorithm can reach.

3. LPA-MNI+ML

3.1 What changes

Only Phase 1 changes. Phase 2 is the same code as in LPA-MNI. In place of one flat pass, Phase 1 is repeated on progressively coarser graphs:

```
1. Run the local-moving pass on the current graph (nodes visited by degree, each moved to the neighboring community with the best modularity gain).
2. Collapse every community into one super-node. Edges between super-nodes carry the summed inter-community weight, and the summed intra-community weight becomes a self-loop so each super-node keeps its original total degree.
3. Repeat on the coarser graph until a pass merges nothing or everything collapses into one community.
4. Map the final super-nodes back to the original nodes and use them as the starting labels for Phase 2.
```

3.2 Why it should help

Because super-nodes can themselves move and merge, the algorithm can combine whole communities, which the flat pass cannot. This is the same mechanism that lets Louvain reach high modularity. Phase 2 then refines the boundaries with label propagation.

3.3 The trade-off

A partition with higher modularity can drift away from a small ground-truth clustering. This is the resolution-limit behavior of modularity-based methods. On Karate Club the multi-level version finds 3 communities, where the real split has 2, and NMI falls from 0.837 to 0.691.

3.4 One important observation about novelty

Phase 1 of LPA-MNI+ML is essentially Louvain. So any claim that the algorithm is a contribution beyond Louvain rests on what Phase 2 adds on top of it. The current results do not isolate that effect. An ablation (Louvain alone, Phase 1 alone, Phase 2 alone, both) is the experiment that would.

4. Experimental setup

4.1 Datasets

Dataset	Size	Ground truth	Reliability
Karate Club	34 nodes, 78 edges	Real 2-faction split	Established
Dolphins	62 nodes, 159 edges	2-way split from Girvan-Newman (21 vs 41)	Proxy, computed from the graph's own edges, so NMI is circular
Florentine Families	15 nodes, 20 edges	Medici vs oligarch parties (Padgett and Ansell)	Proxy, some allegiances debated
American College Football	115 nodes, 613 edges	Real conference (12)	Established
Cora	2708 nodes, 5278 edges	Paper topic (7)	Established, but topics need not match graph structure

Dataset	Size	Ground truth	Reliability
Yeast	1458 nodes, 1948 edges	Girvan-Newman dendrogram cut at max modularity (39 communities)	Proxy, computed from the graph's own edges, so NMI is circular

Newman's Dolphins file contains only names. Lusseau's field-observed groups are not in it, so the ground truth used here is a stand-in and should be replaced before any published use. The Yeast protein-interaction network comes from Network Repository (Rossi and Ahmed, AAAI 2015). Its file has anonymized node IDs and no functional labels, so its ground truth is also derived from the graph and would be better replaced by real annotations such as MIPS or GO classes.

4.2 Algorithms and metrics

Five algorithms are compared: LPA, LPA-MNI, FLPA, LPA-S, LPA-MNI+ML. Two metrics score each partition. **Modularity** needs no labels and measures density against a random graph, so it favors algorithms that optimize it. **NMI** measures agreement with the ground-truth partition and is neutral about method, but is only as reliable as the labels. Edge weights are ignored throughout, as the algorithms are defined for unweighted graphs.

5. Results

5.1 Modularity (bold = best in row)

Dataset	LPA	LPA-MNI	FLPA	LPA-S	LPA-MNI+ML
Karate Club	0.309	0.404	0.417	0.404	**0.435**
Dolphins	0.499	0.503	0.502	0.511	**0.511**
Florentine Families	0.335	0.289	0.351	0.289	**0.391**
American College Football	0.583	0.557	0.554	0.581	**0.605**
Cora	0.652	0.726	0.695	0.710	**0.816**
Yeast	0.630	0.727	0.690	0.756	**0.818**

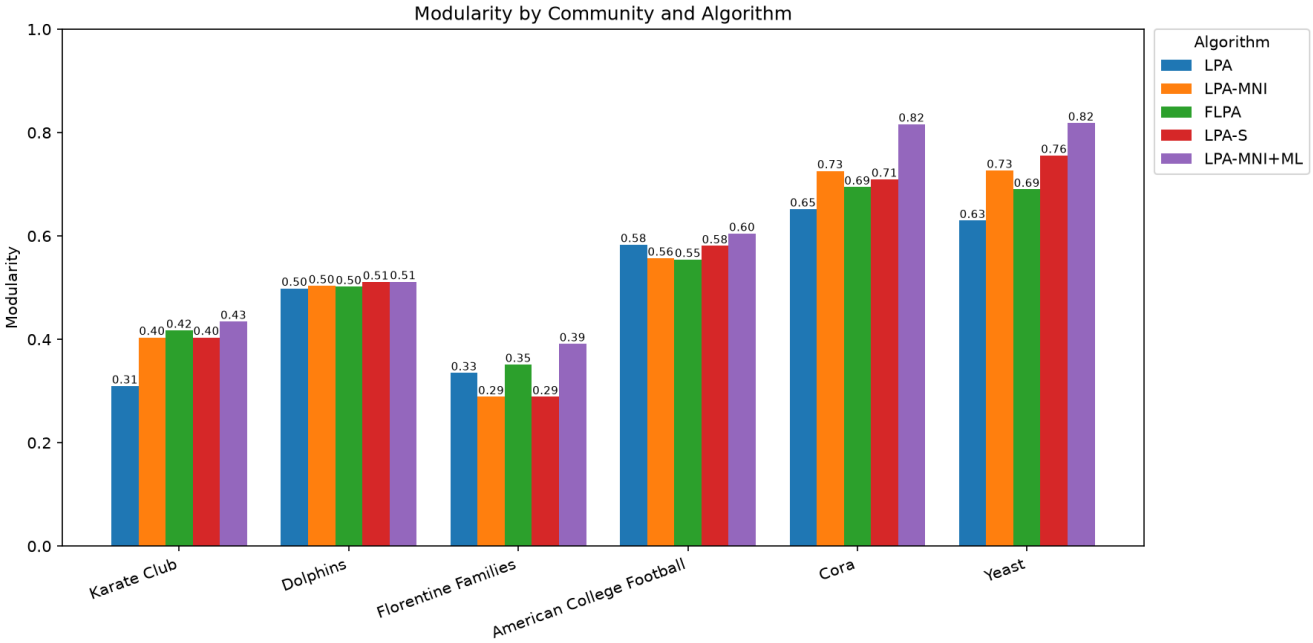
5.2 NMI (bold = best in row)

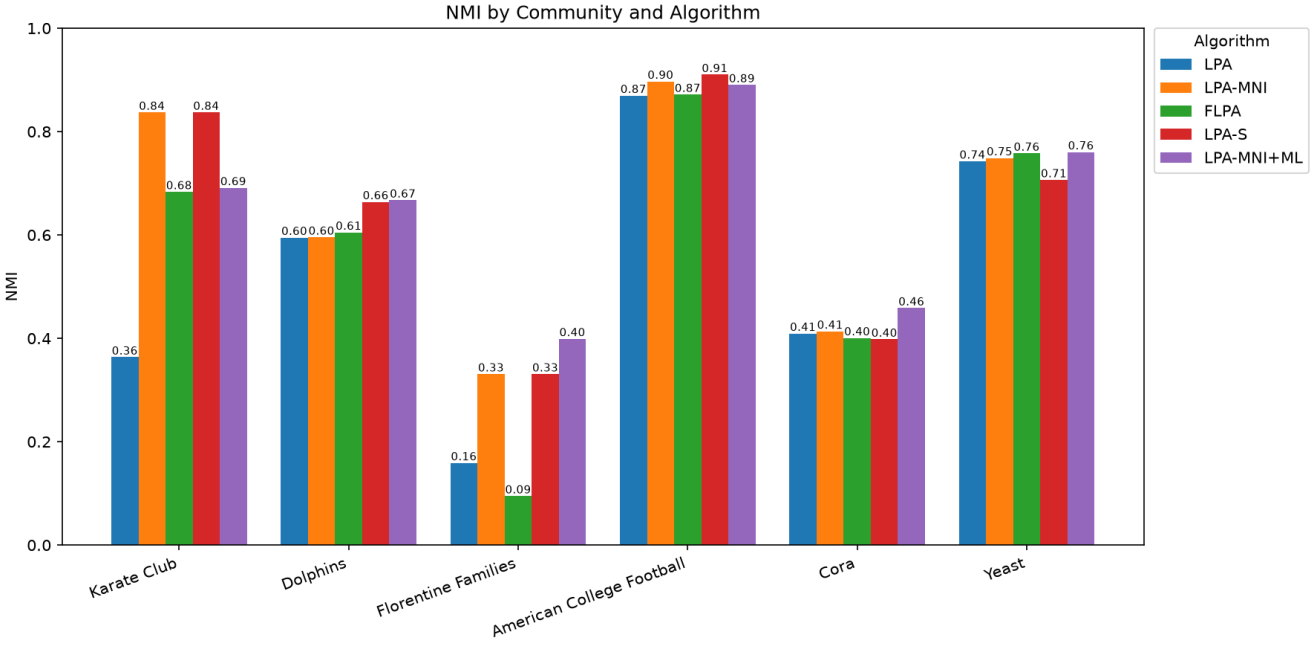
Dataset	LPA	LPA-MNI	FLPA	LPA-S	LPA-MNI+ML
Karate Club	0.364	**0.837**	0.683	**0.837**	0.691
Dolphins*	0.595	0.596	0.605	0.664	**0.667**
Florentine Families*	0.159	0.331	0.095	0.331	**0.399**
American College Football	0.870	0.897	0.872	**0.910**	0.890
Cora	0.409	0.414	0.401	0.398	**0.459**
Yeast*	0.743	0.749	0.758	0.707	**0.760**

* NMI uses a proxy ground truth for these datasets. See section 4.1.

5.3 Charts

The two charts below show the same five algorithms as the tables above.


Modularity by Community and Algorithm
Modularity
Algorithm
LPA
LPA-MNI
FLPA
LPA-S
LPA-MNI+ML
Karate Club
Dolphins
Florentine Families
American College Football
Cora
Yeast


NMI by Community and Algorithm
NMI
Algorithm
LPA
LPA-MNI
FLPA
LPA-S
LPA-MNI+ML
Karate Club
Dolphins
Florentine Families
American College Football
Cora
Yeast

6. What the results support

6.1 Multi-level seeding raises modularity

Dataset	LPA-MNI	LPA-MNI+ML	Change	Lead over next best
Karate Club	0.404	0.435	+0.031	+0.0172
Dolphins	0.503	0.511	+0.008	+0.0005
Florentine Families	0.289	0.391	+0.103	+0.0400
American College Football	0.557	0.605	+0.048	+0.0214
Cora	0.726	0.816	+0.090	+0.0904
Yeast	0.727	0.818	+0.092	+0.0628

Both algorithms are deterministic, so these gains are not seed noise. The change is large on Cora, Yeast, Florentine and Football, and small on Dolphins and Karate. LPA-MNI+ML has the highest modularity of the five on 6 of 6 datasets. Its lead over the next best algorithm is large on Cora (0.090), Yeast (0.063) and Florentine (0.040), but only 0.0005 on Dolphins, which is within noise. Because modularity-optimizing methods are not in this comparison, these leads say nothing about how LPA-MNI+ML compares with them.

6.2 The NMI effect is mixed

Dataset	LPA-MNI	LPA-MNI+ML	Change	Best in row
Karate Club	0.837	0.691	-0.146	LPA-MNI (0.837)
Dolphins*	0.596	0.667	+0.072	LPA-MNI+ML (0.667)
Florentine Families*	0.331	0.399	+0.068	LPA-MNI+ML (0.399)
American College Football	0.897	0.890	-0.007	LPA-S (0.910)
Cora	0.414	0.459	+0.045	LPA-MNI+ML (0.459)
Yeast*	0.749	0.760	+0.011	LPA-MNI+ML (0.760)

NMI improves on Florentine, Dolphins and Cora, improves slightly on Yeast, is nearly unchanged on Football, and drops on Karate Club. The Karate drop is the clearest instance of the trade-off in section 3.3. The one dataset where multi-level seeding hurts NMI is a small graph with a very coarse true structure (two factions). LPA-MNI+ML has the highest NMI of the five on Dolphins, Florentine Families, Cora, Yeast. Three of those (Dolphins, Florentine Families, Yeast) use ground truth that was constructed rather than observed, and on Dolphins and Yeast it was derived from the graph itself, so these NMI leads carry little weight.

6.3 Higher modularity does not mean higher NMI

Higher modularity does not reliably give higher NMI. On Karate Club, LPA-MNI+ML has higher modularity than LPA-MNI (0.435 vs 0.404) but much lower NMI (0.691 vs 0.837). On Football, LPA-MNI+ML has the highest modularity (0.605) but LPA-S has the higher NMI (0.910 vs 0.890). Reporting only one metric would tell a different story, so both should always be shown.

6.4 No algorithm wins everywhere

Even within this set of five, the best NMI changes from dataset to dataset: LPA-MNI on Karate Club and LPA-S on Football are ahead of LPA-MNI+ML. Plain LPA has the lowest modularity on Karate Club, Dolphins, Cora,

Yeast. With six datasets this is a description of the benchmark, not a general ranking.

6.5 Yeast: NMI can reward fragmentation

In the Yeast run, the proxy ground truth has 39 communities. Plain LPA finds 334 communities and still scores NMI 0.743, close to LPA-MNI+ML (0.760), which finds 31. NMI is known to be biased upward when a partition has many small clusters, so Yeast NMI values barely separate these algorithms and should not be compared at face value. On modularity the picture is cleaner: LPA-MNI+ML reaches 0.8184 against 0.6298 for plain LPA, and multi-level seeding lifts LPA-MNI by 0.092. Because the proxy truth itself maximizes modularity (Q = 0.814), partitions with a similar Q agree with it by construction. Community counts quoted here are from the run that produced the results tables.

7. What the results do not support

- **That LPA-MNI+ML beats existing methods in general.** This report compares five label-propagation algorithms only. Being best of five says nothing about how it compares with modularity optimizers or other families that are outside this comparison, and the NMI leads are mostly on datasets with constructed ground truth.
- **Rankings among close algorithms.** Gaps of about 0.01 to 0.02 are inside the noise. LPA-MNI on Cora changed from 266 to 270 communities when only the edge insertion order changed, because the pseudocode leaves some tie-breaks open.
- **Anything precise about Karate Club or Florentine Families.** At 34 and 15 nodes, moving one or two nodes shifts NMI by a lot.
- **That Phase 2 contributes anything beyond Louvain.** No experiment isolates it yet.
- **Generalization.** Six datasets, one run each, cannot support a claim about graphs in general.

A defensible statement today is: *multi-level seeding substantially improves LPA-MNI's modularity, at a modest or mixed cost in NMI.* The numbers support that. 'LPA-MNI+ML outperforms existing methods' is not yet supported.

8. Recommended next steps

Step	Why	Changes the algorithm?
Repeated runs (30+ seeds), report mean and standard deviation	Separates real differences from noise. Everything else depends on this.	No
Ablation: Louvain alone, Phase 1 only, Phase 2 only, both	Shows whether Phase 2 adds anything beyond Louvain, which is the core novelty question.	No
Leiden-style connectivity check	Fixes disconnected communities, a known weakness of Louvain-style seeding.	Yes
Phase 2 refinement between coarsening levels	Lets early mislabeled nodes be corrected before they are merged.	Yes
Resolution parameter chosen without ground truth	Addresses the Karate over-merging. Must not be tuned on NMI.	Yes
Replace proxy labels (use Lusseau's observed Dolphin groups)	Removes the circularity in the Dolphins NMI.	No

On tuning. If settings are adjusted until NMI rises on these six datasets, the result is fitted to the test set. Select the method by label-free criteria, then report NMI once.

Appendix: implementation notes

- **LPA-MNI** was checked against the supplied pseudocode and matches it. Nodes are ordered by degree (equivalent to DC ordering), Phase 1 moves a node only on a strictly positive modularity gain, and Phase 2 is asynchronous with deterministic DC tie-breaking. A weighted path exists in Phase 1 but only Karate carries weights, and its result is identical without them.
- **LPA-S** was first implemented from an earlier, simpler pseudocode in which each node copies its single best neighbor. It was then rewritten to match the final pseudocode, in which similarity is summed per label.
- **MFLP** was implemented twice from the paper (once from the prose, once from supplied pseudocode) and did not reproduce the paper's reported scores. It was removed from the comparison, and its file remains in `algorithms/mflp.py`.
- **Excluded from this report:** three further algorithms exist in the project but their results are left out of this report by request, and so is a dataset used earlier.
- **Dolphins ground truth** is the first Girvan-Newman two-way split. It is derived from the graph and is therefore a proxy.
- **Yeast** comes from Network Repository (bio-yeast). Its ground truth is the Girvan-Newman edge-betweenness dendrogram cut at its maximum-modularity level (39 communities, Q = 0.814), computed with igraph. It is derived from the graph, so it is a proxy and circular.